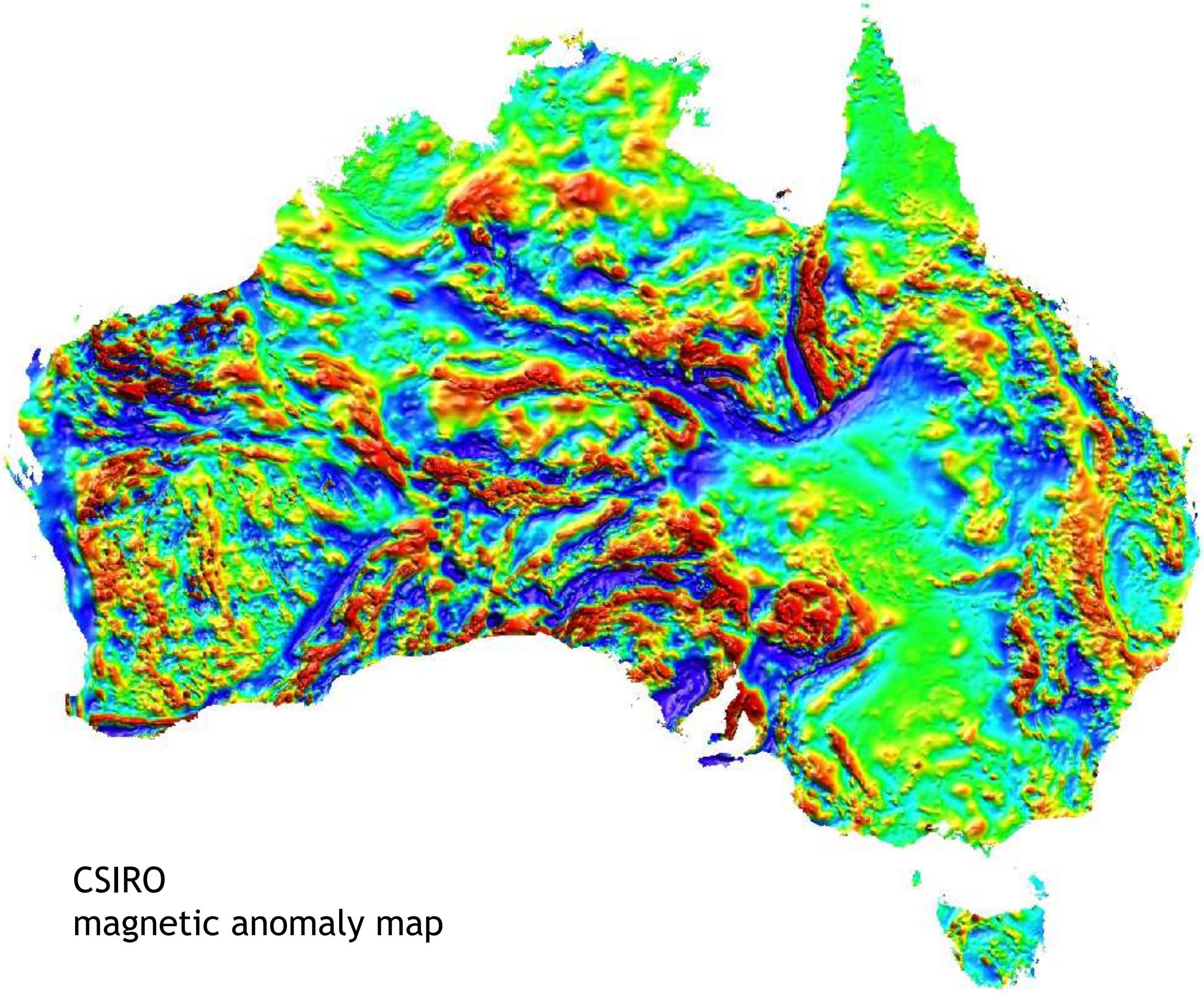
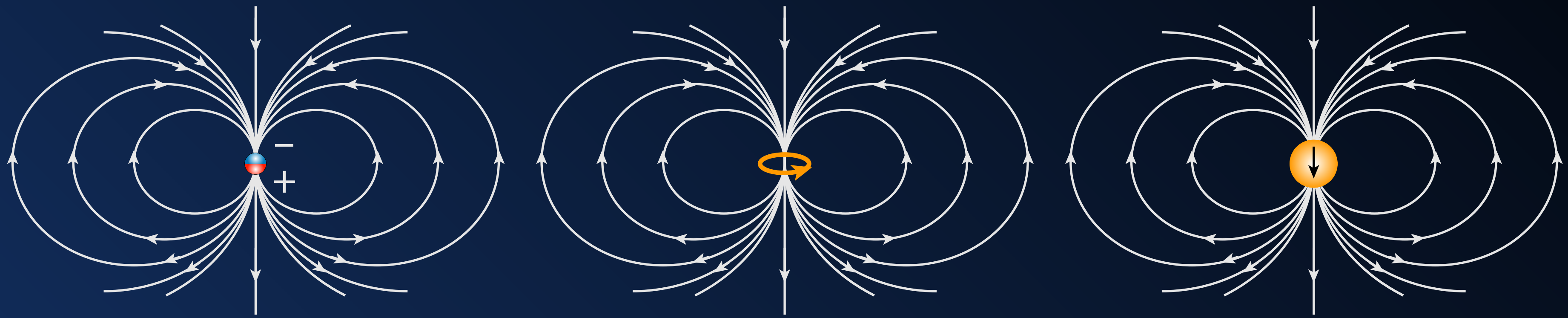
Geophysics III
Magnetics



CSIRO
magnetic anomaly map

What causes magnetic fields?
Three ways to conceptulize the magnetic field

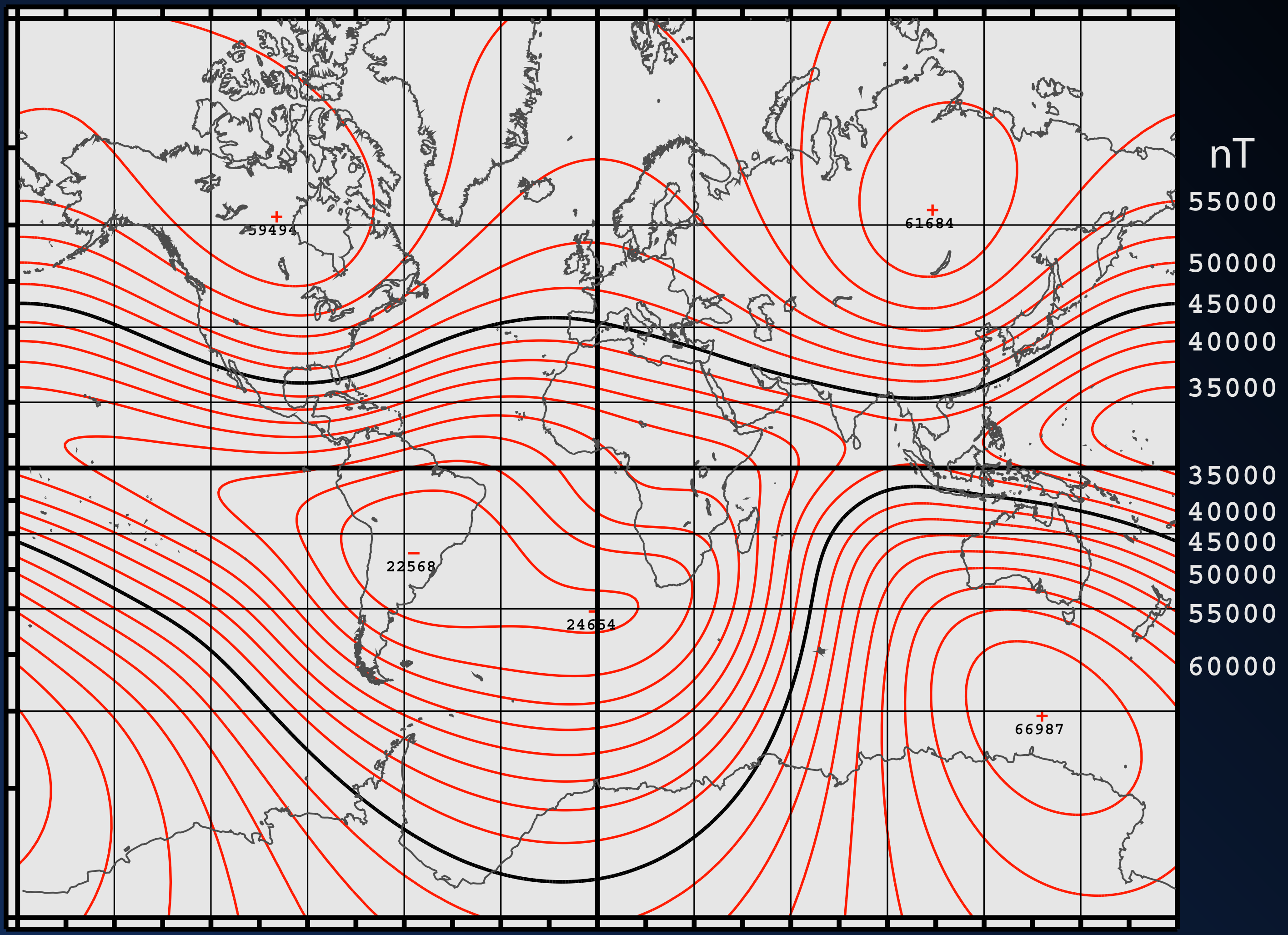


magnetic poles (charges)

loop currents

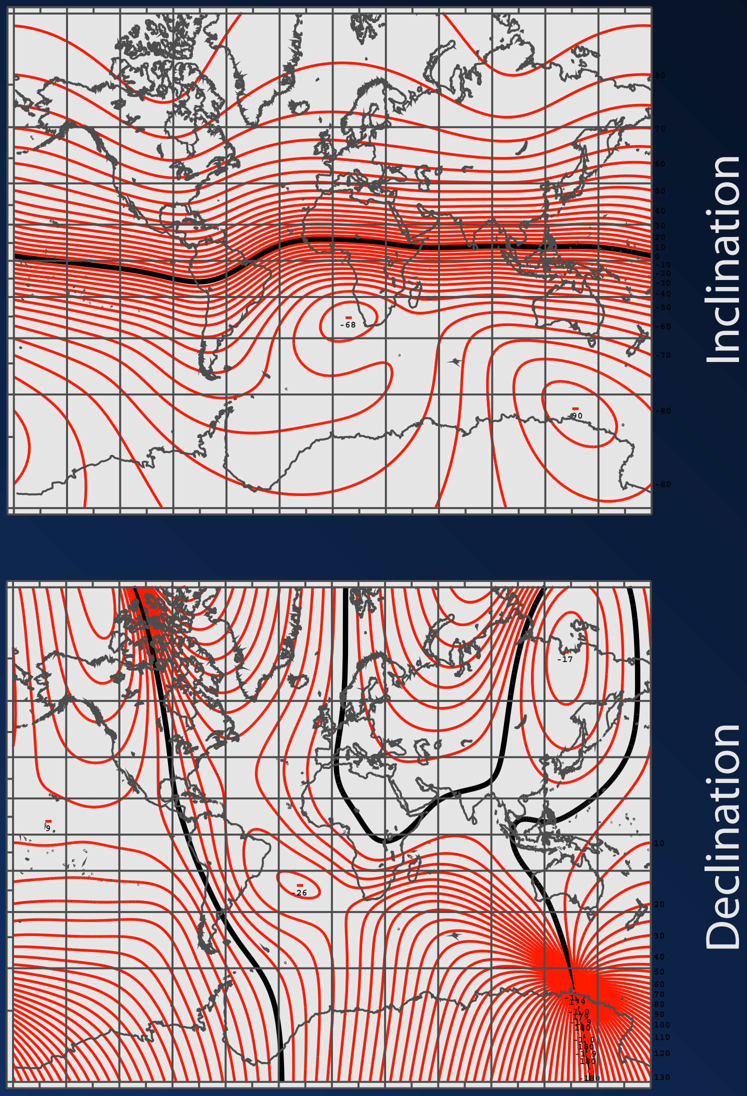
uniformly magnetized body

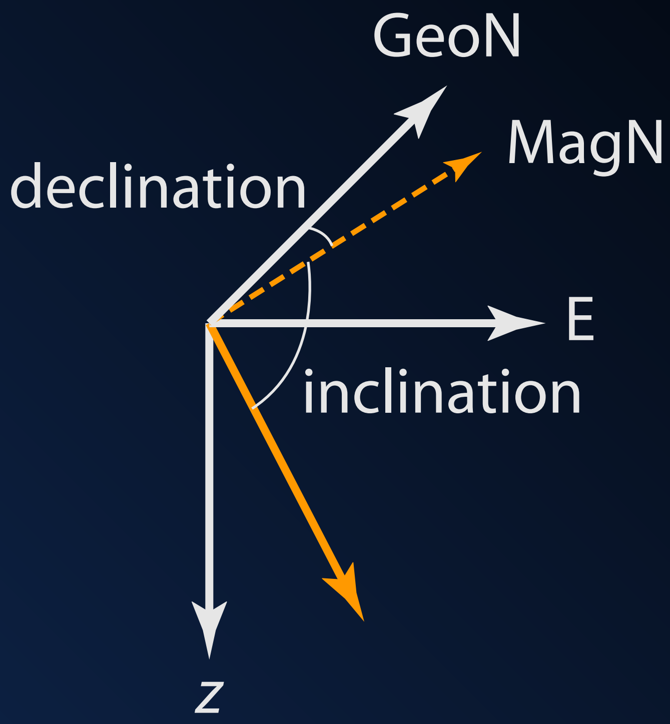
Earth's Core Field (2010)



IGRF - Total Intensity

Earth’s Core Field (2010)





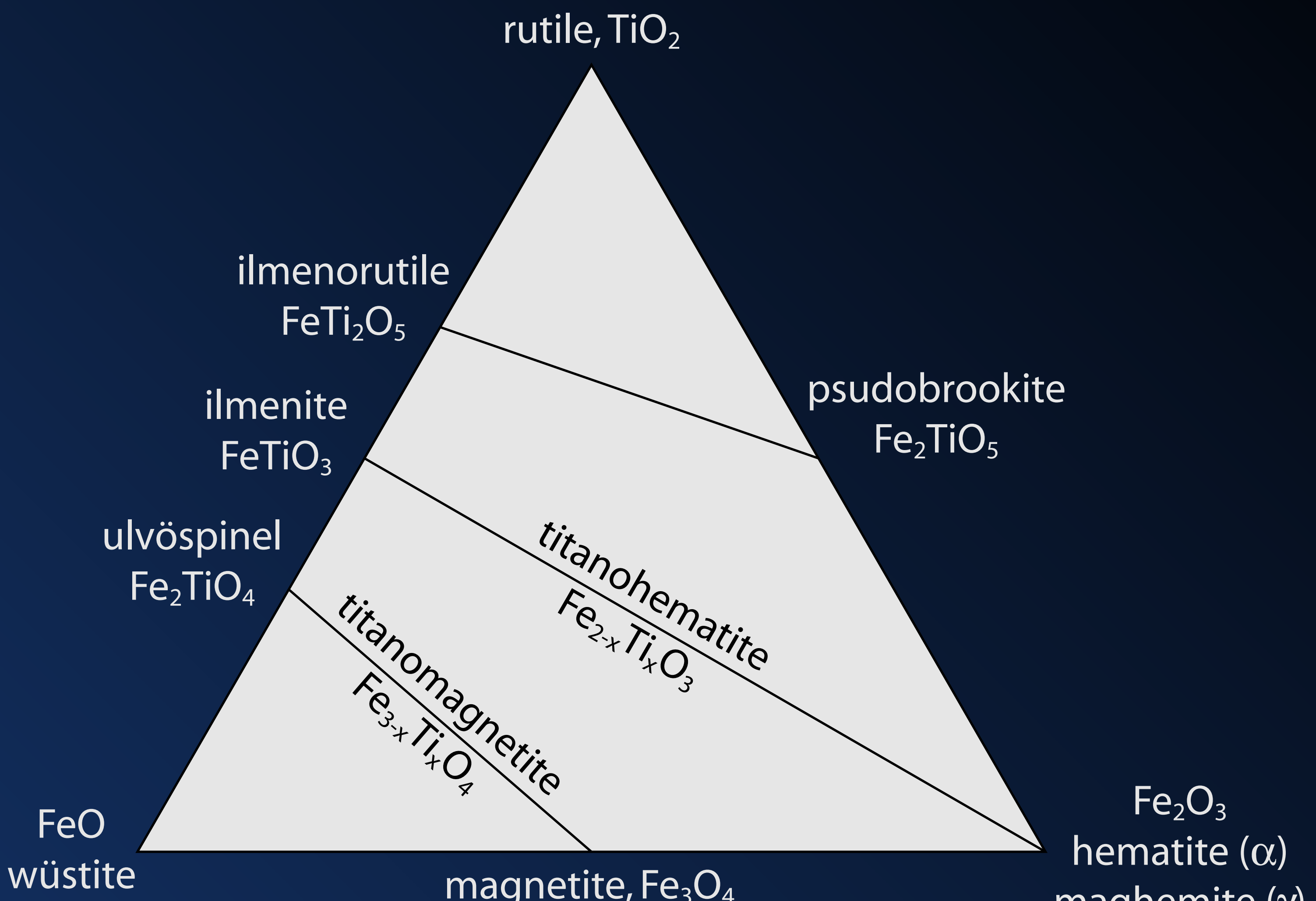
Magnetic Susceptibility of Common Rocks

Rock Type	Susceptibility Range
limestone	10 to 25000
sandstone	0 to 21000
shale	60 to 18600
schist	315 to 3000
rhyolite	250 to 37700
granite	10 to 65
granite (w/ magnetite)	20 to 50000
pegmatite	3000 to 75000
basalt	500 to 182000
gabbro	800 to 76000
peridotite	95500 to 196000

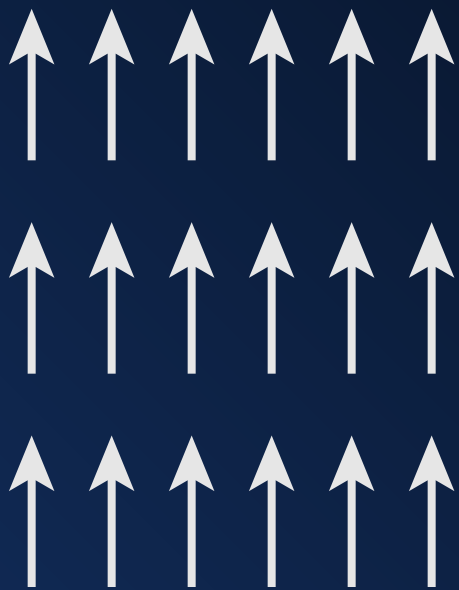
Increasing with mafic content

From Reynolds (1997) pg. 127

Not all minerals are magnetic.
But Fe-Ti oxides are!

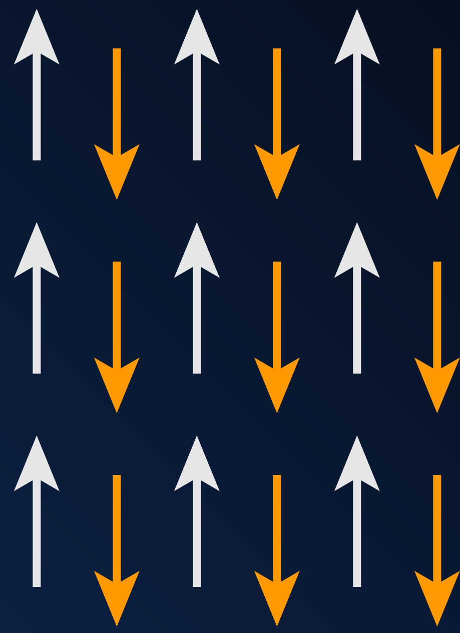


Three types of ferromagnetism



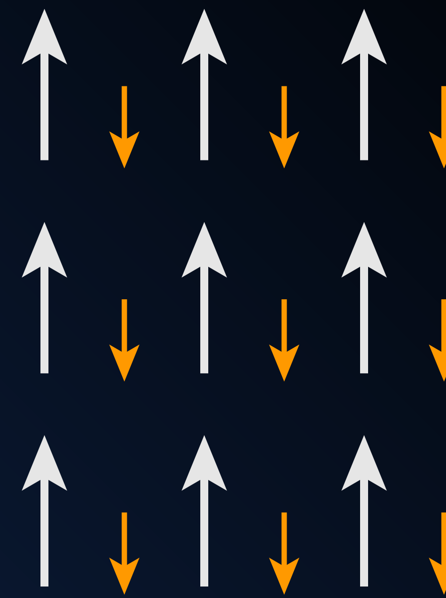
Ferromagnetic

iron, nickel,
cobalt
goetite (weakly)



Antiferromagnetic

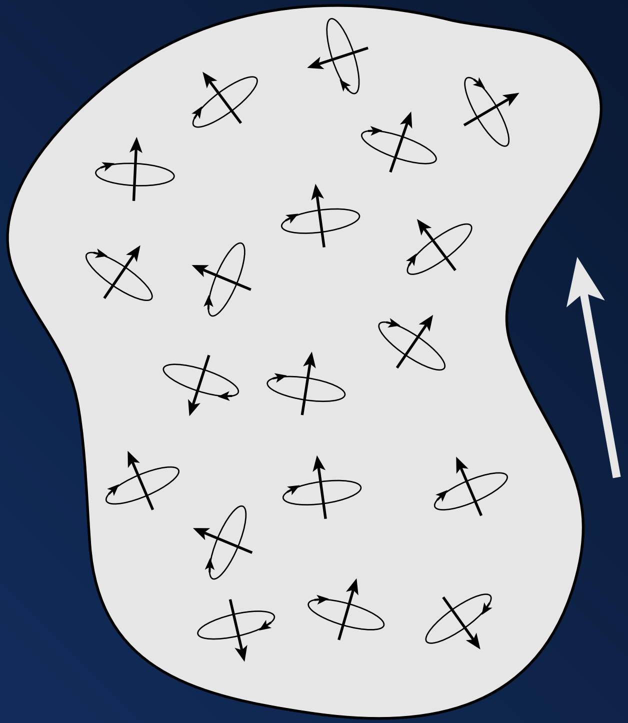
hematite
titanohematite ($0 < x < 0.45$)



Ferrimagnetic

magnetite
maghemite
titanomagnetite
titanohematite ($x > 0.45$)

What is Magnetization? Magnetization distorts the magnetic field.



$$\mathbf{M} = \frac{1}{V}\sum_{i=1}^{N} \mathbf{m}_i$$

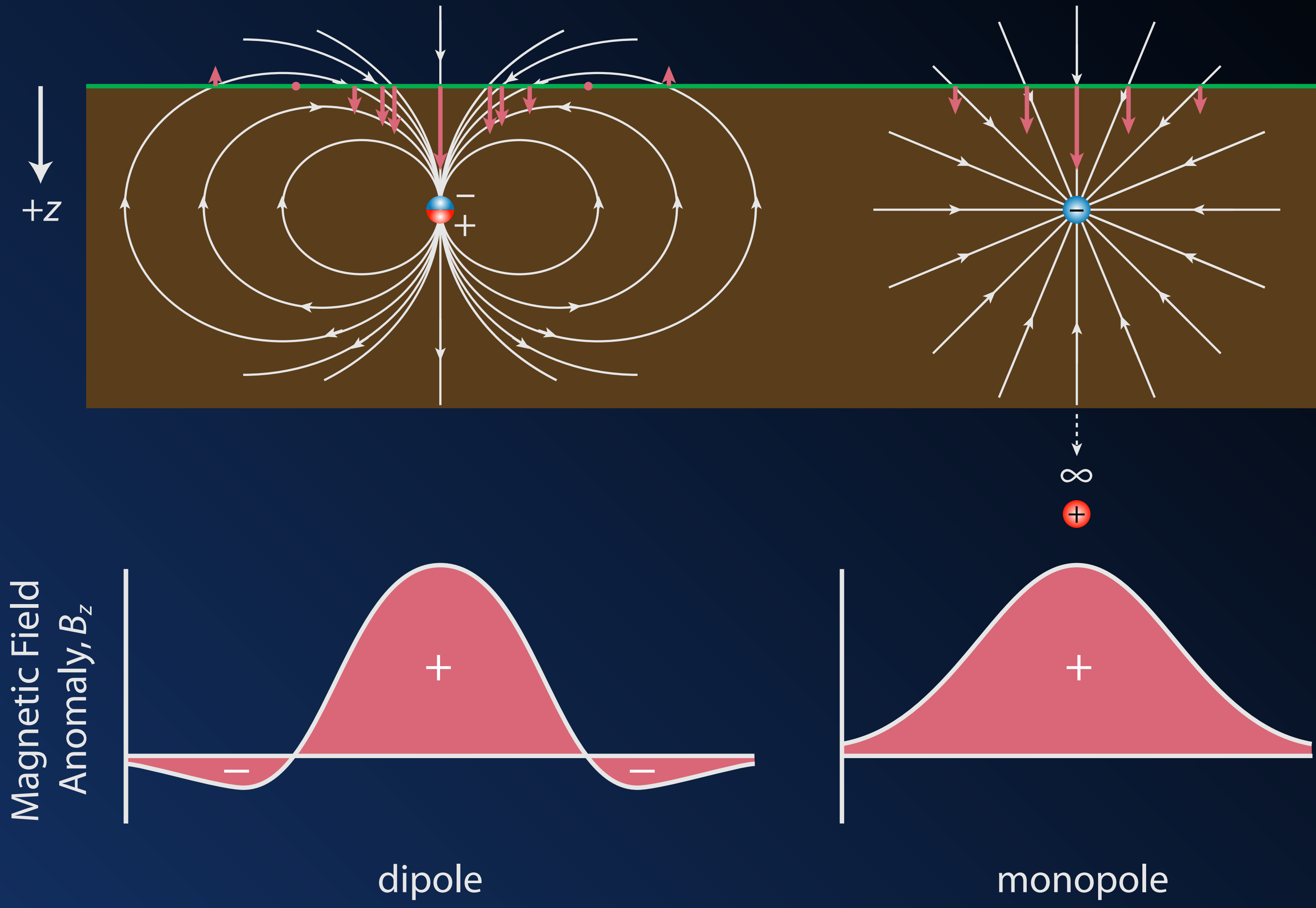
$$\mathbf{H} = \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}$$

$$\mathbf{M} = \chi \mathbf{H}$$

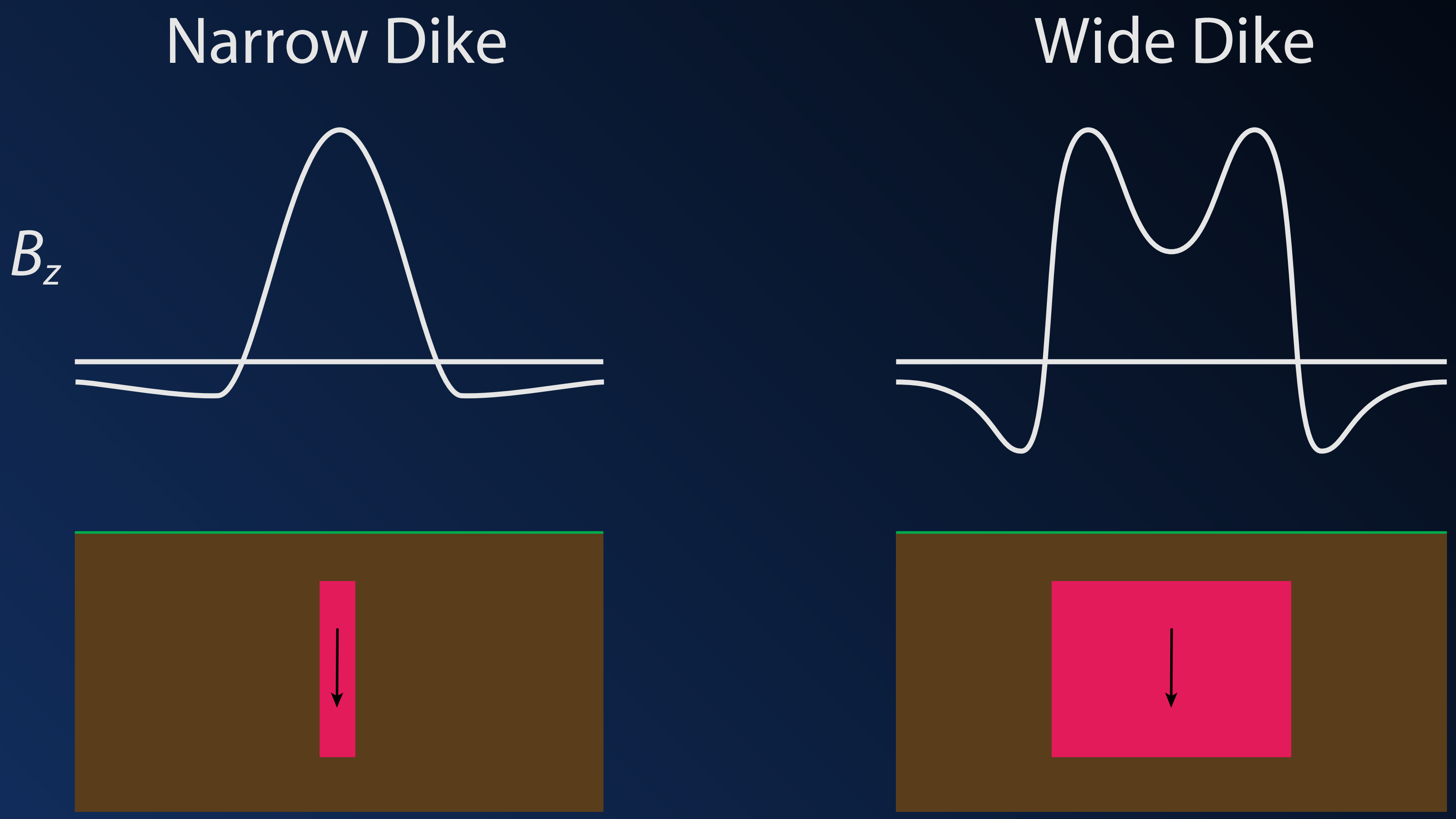
$$\mathbf{B} = \mu_0(1+\chi)\mathbf{H}$$
$$= \mu \mathbf{H}$$

Total $\mathbf{M}$ = induced $\mathbf{M}$ + Remnant $\mathbf{M}$

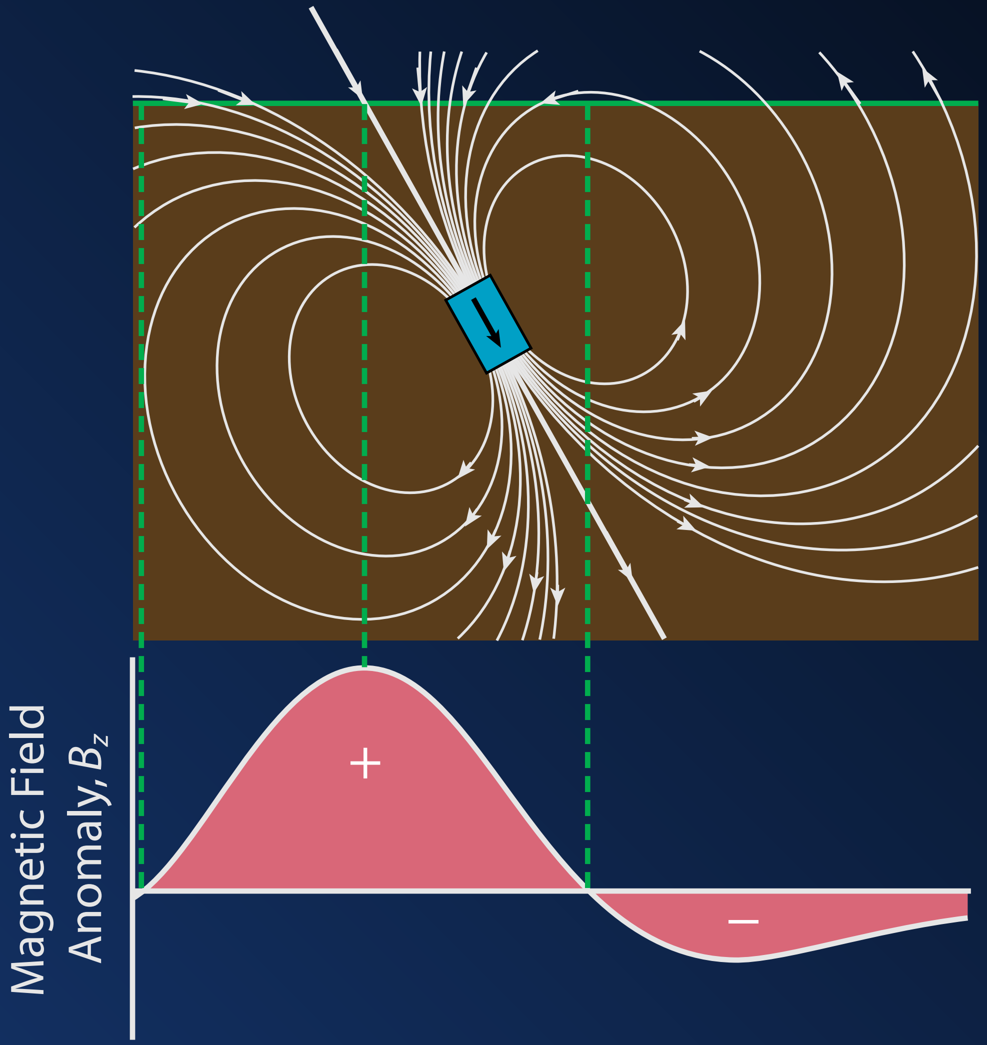
Dipoles and Monopoles



The shape of an anomaly matters.



Orientation matters



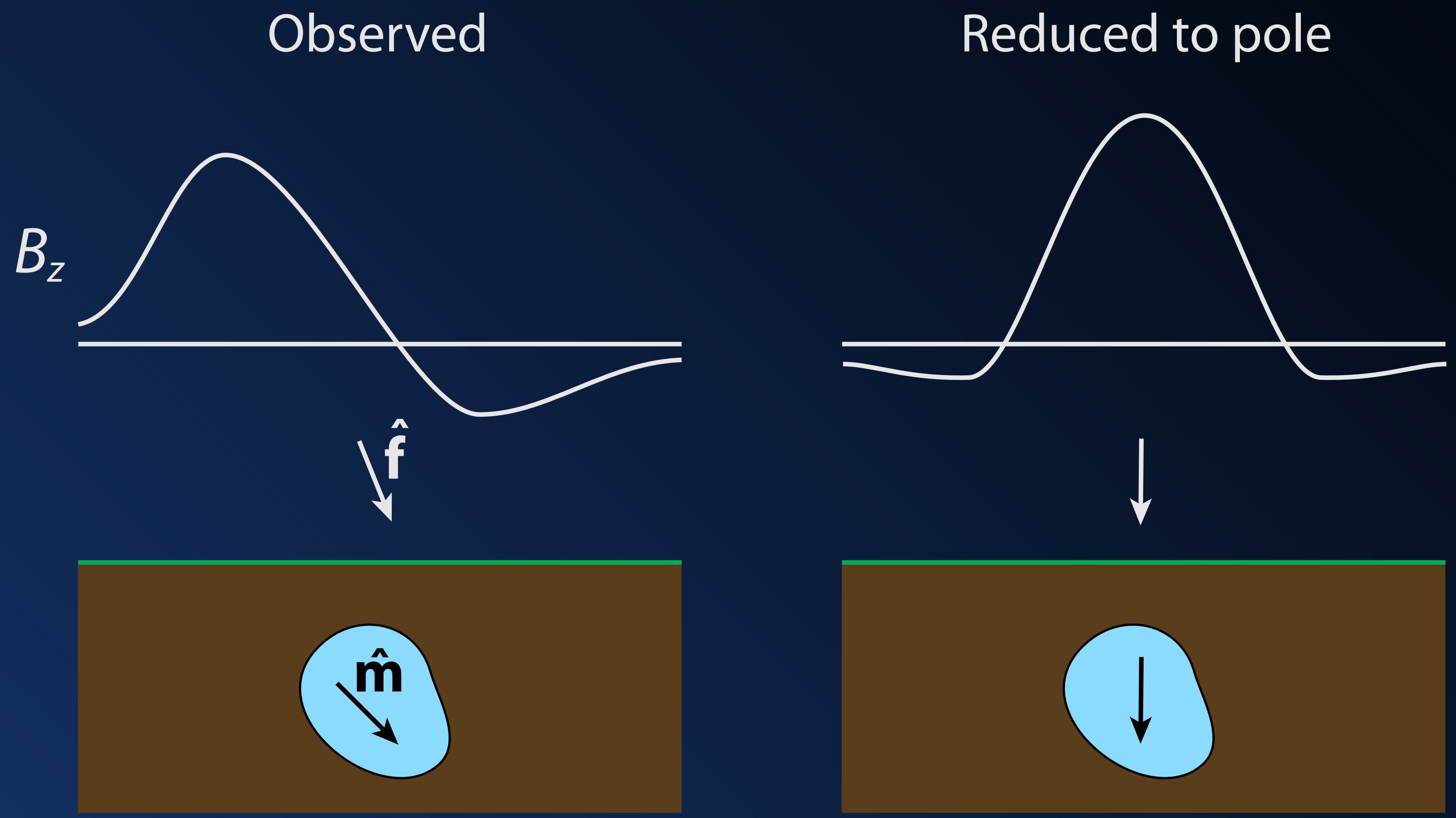
$$B_r = \frac{\mu_0}{2\pi}\frac{m\cos\theta}{r^3}$$

$$B_\theta = \frac{\mu_0}{4\pi}\frac{m\sin\theta}{r^3}$$

$$|B| = \frac{\mu}{4\pi}\frac{m}{r^3}\left[3\cos^2\theta + 1\right]^{1/2}$$

What is Reduction to Pole?

Transforms magnetic fields to vertical.

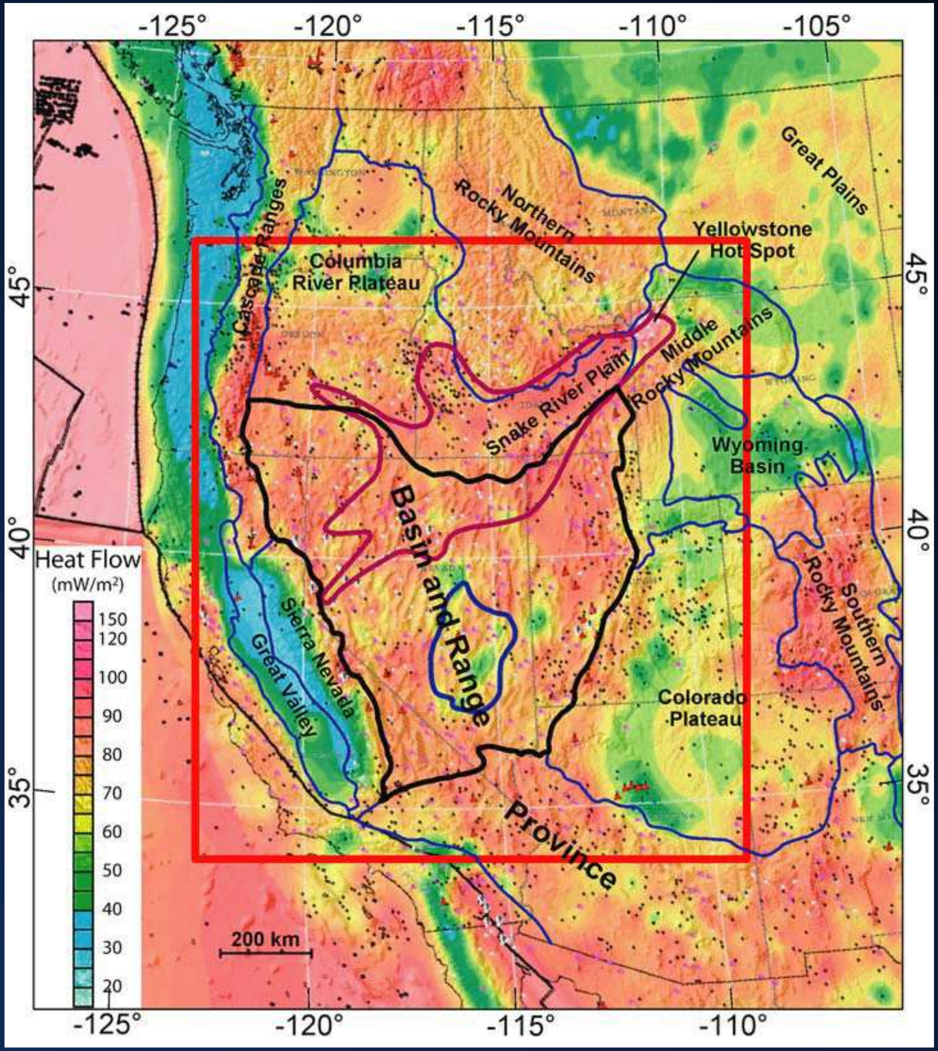


Curie Temperature Sets/Destroys Magnetism

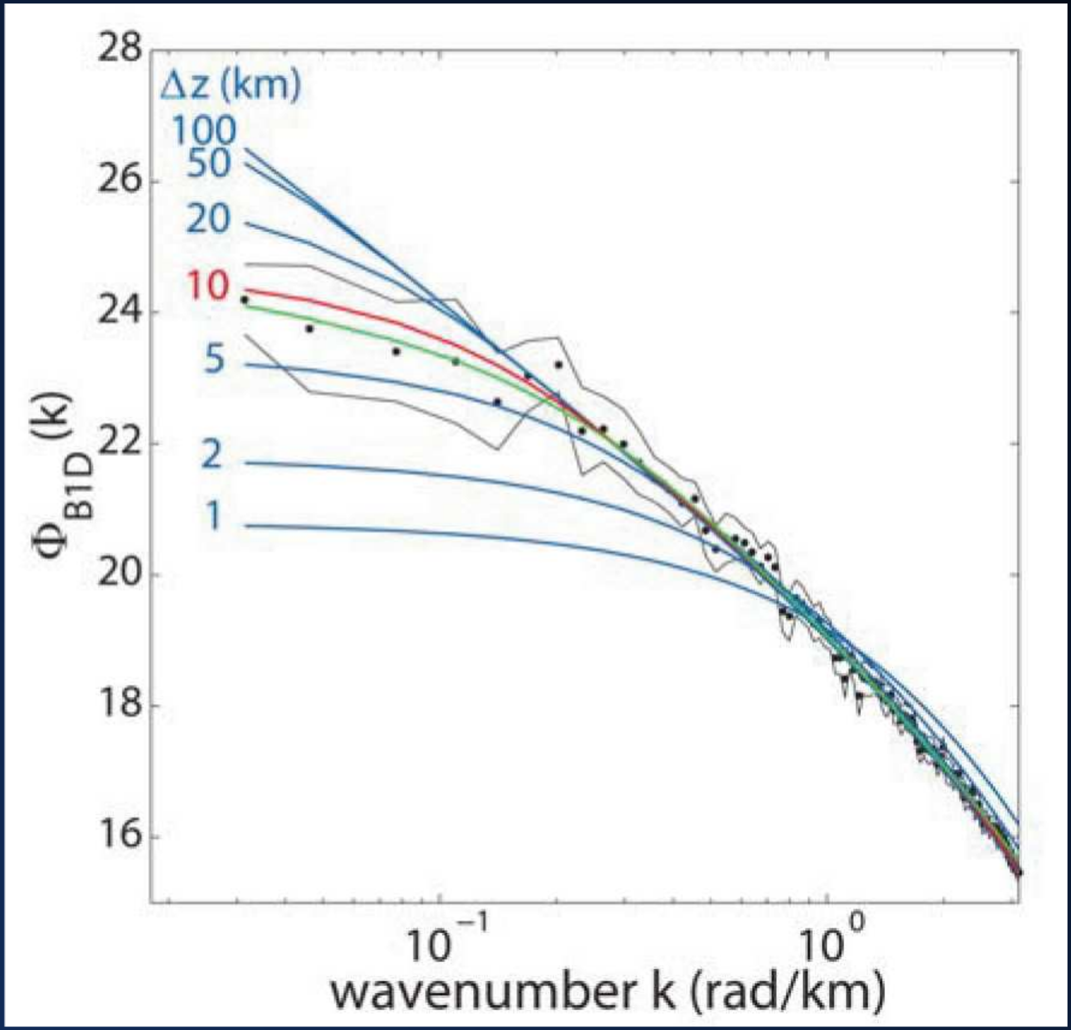
Mineral	Composition	T_C (˚C)
iron	αFe	765
magnetite	Fe_3O_4	580
hematite	αFe_2O_3	675
maghemite	γFe_2O_3	590 to 675
titanomagnetite	$Fe_{2.7}Ti_{0.3}O_4$	350
	$Fe_{2.4}Ti_{0.6}O_4$	150
goethite	$\alpha FeO(OH)$	120
pyrrhotite	Fe_7S_8	320
greigite	Fe_3S_4	~330

From Dunlop and Özdemir (1997) p. 51

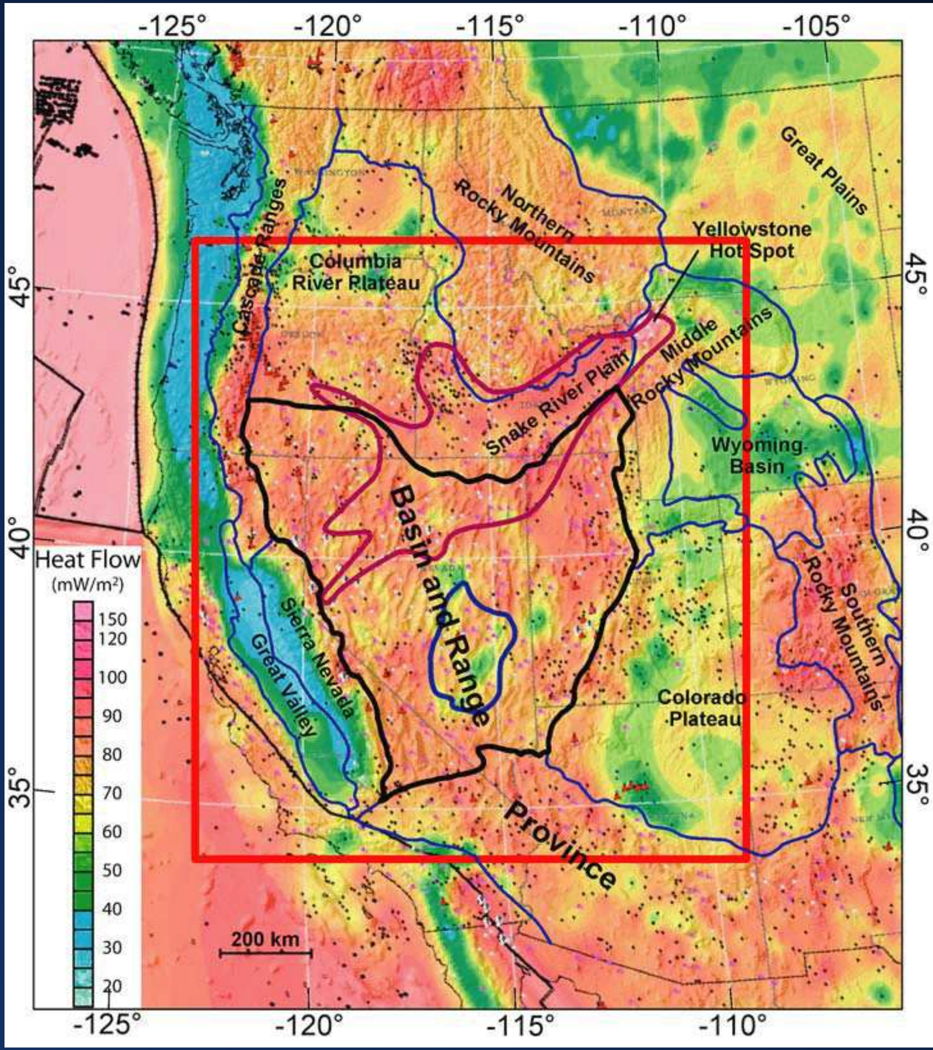
Curie Depth: Case Study
Observed Heat Flow

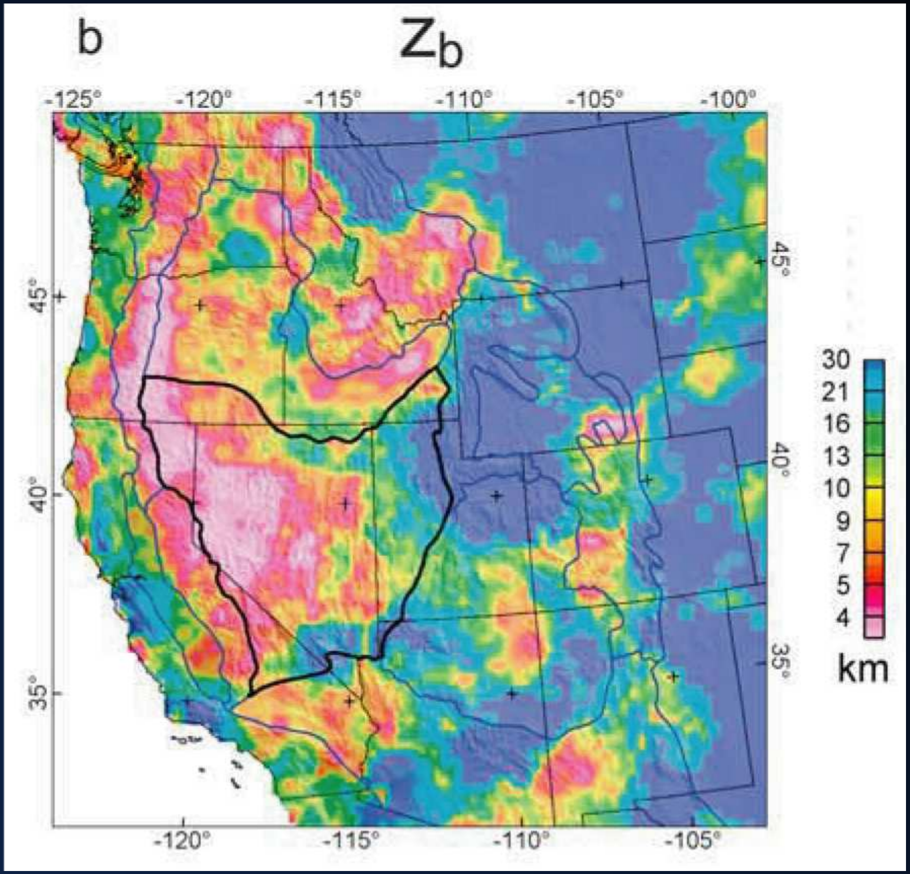


Curie Depth: Case Study
Thickness of Magnetic Crust

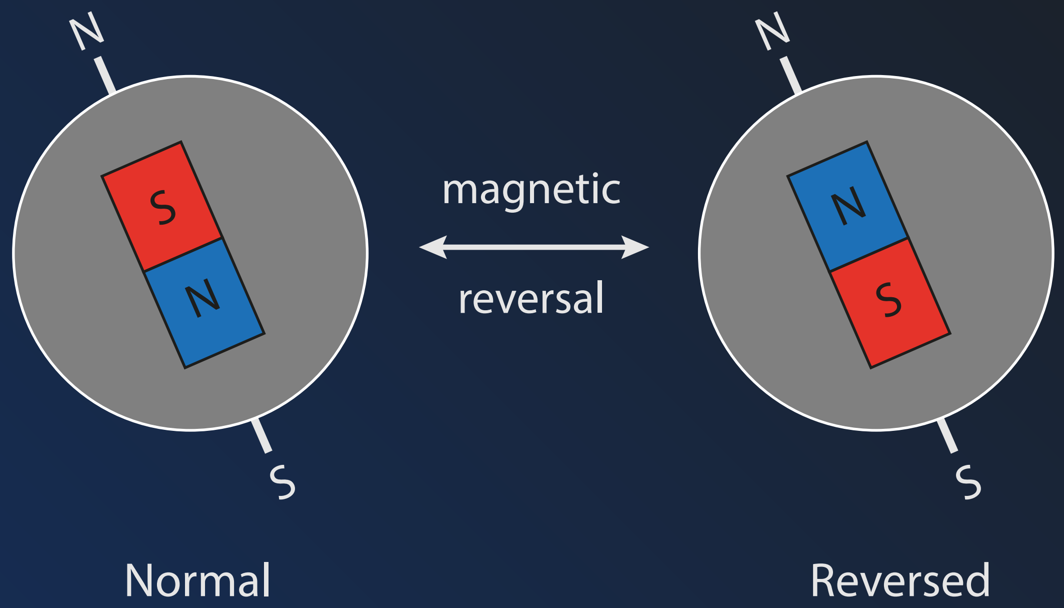


Curie Depth: Case Study
Comparison with heat flow





Geomagnetic Reversal Timescale

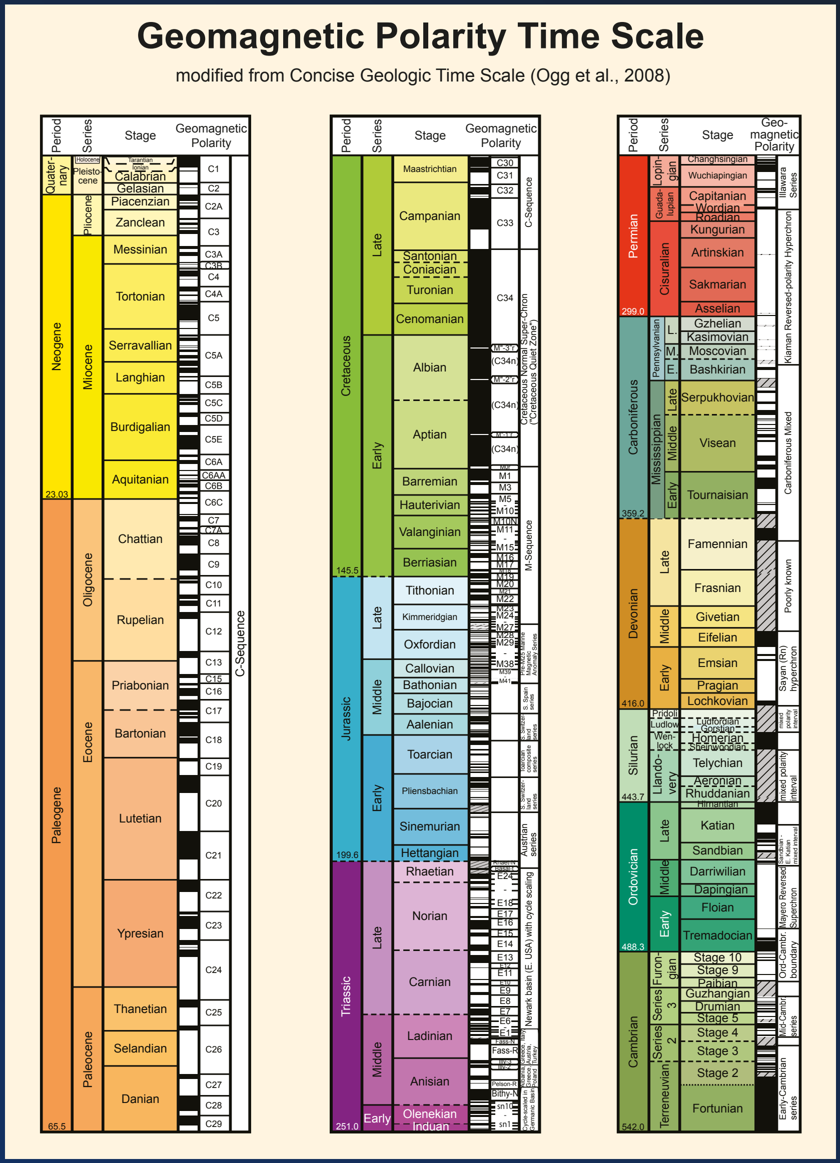


Dynamics in core cause polarity of Earth's magnetic field to flip

Process is chaotic, but reversals tend to occur ~100 ky

*The north pole of a magnet points to the "North" pole of the Earth...
What must the polarity of the Earth's North pole be?

Geomagnetic Reversal Timescale

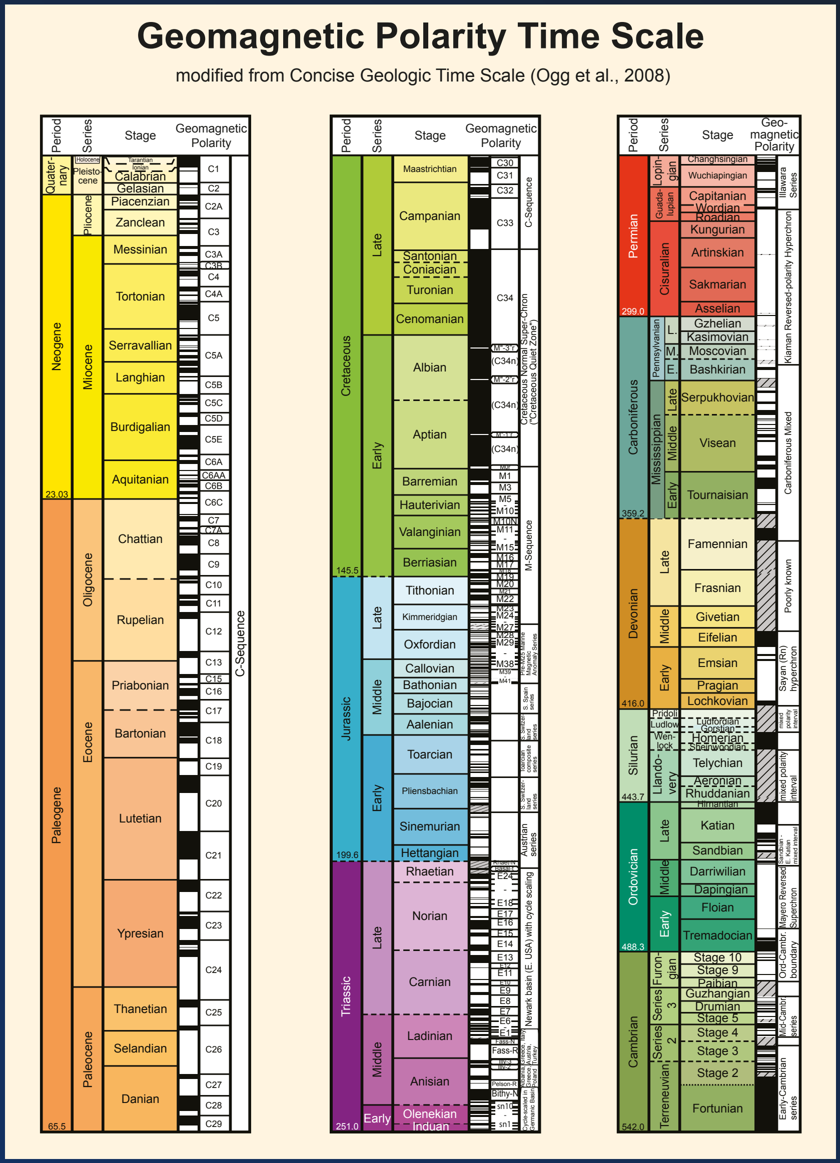


Useful when exact ages are unknown

especially within sediments

Ogg et al., 2010

Geomagnetic Reversal Timescale



Useful when exact ages are unknown

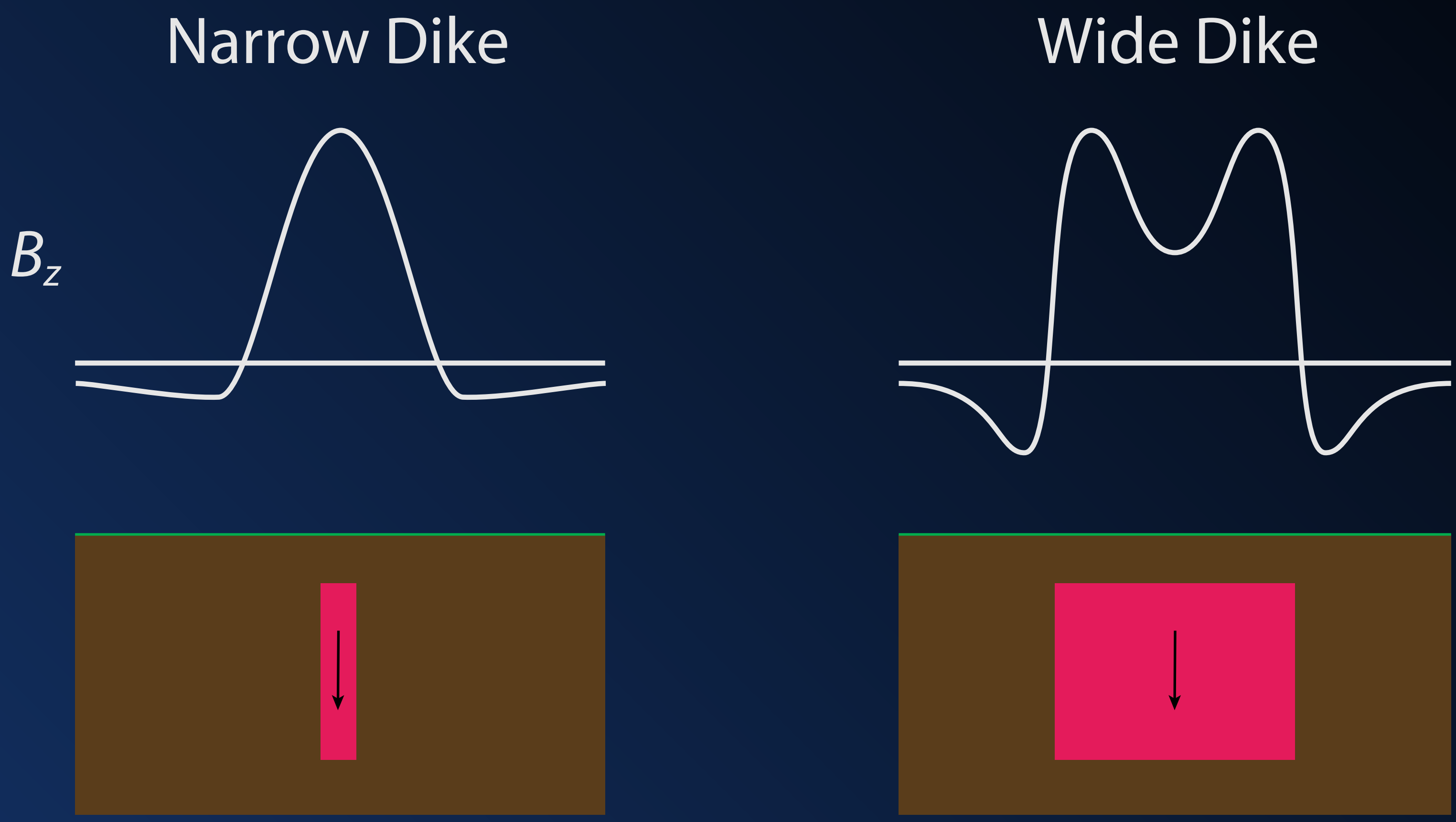
especially within sediments

Used to calibrate other age proxies e.g. pollen, diatoms, isotope records

Used to cross-correlate stratigraphic sections

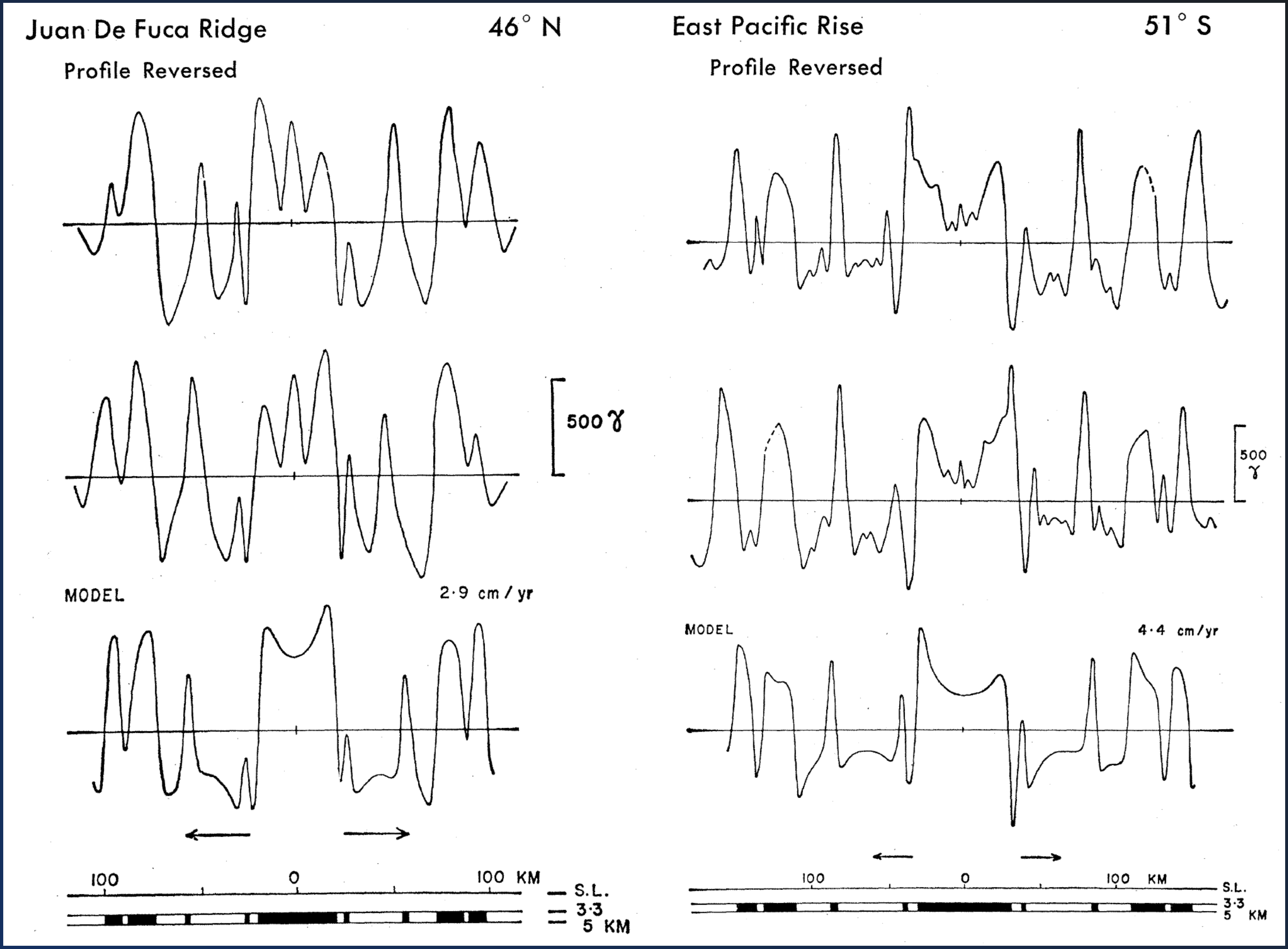
Used to date lava flows on the ocean floor

The shape of an anomaly matters.

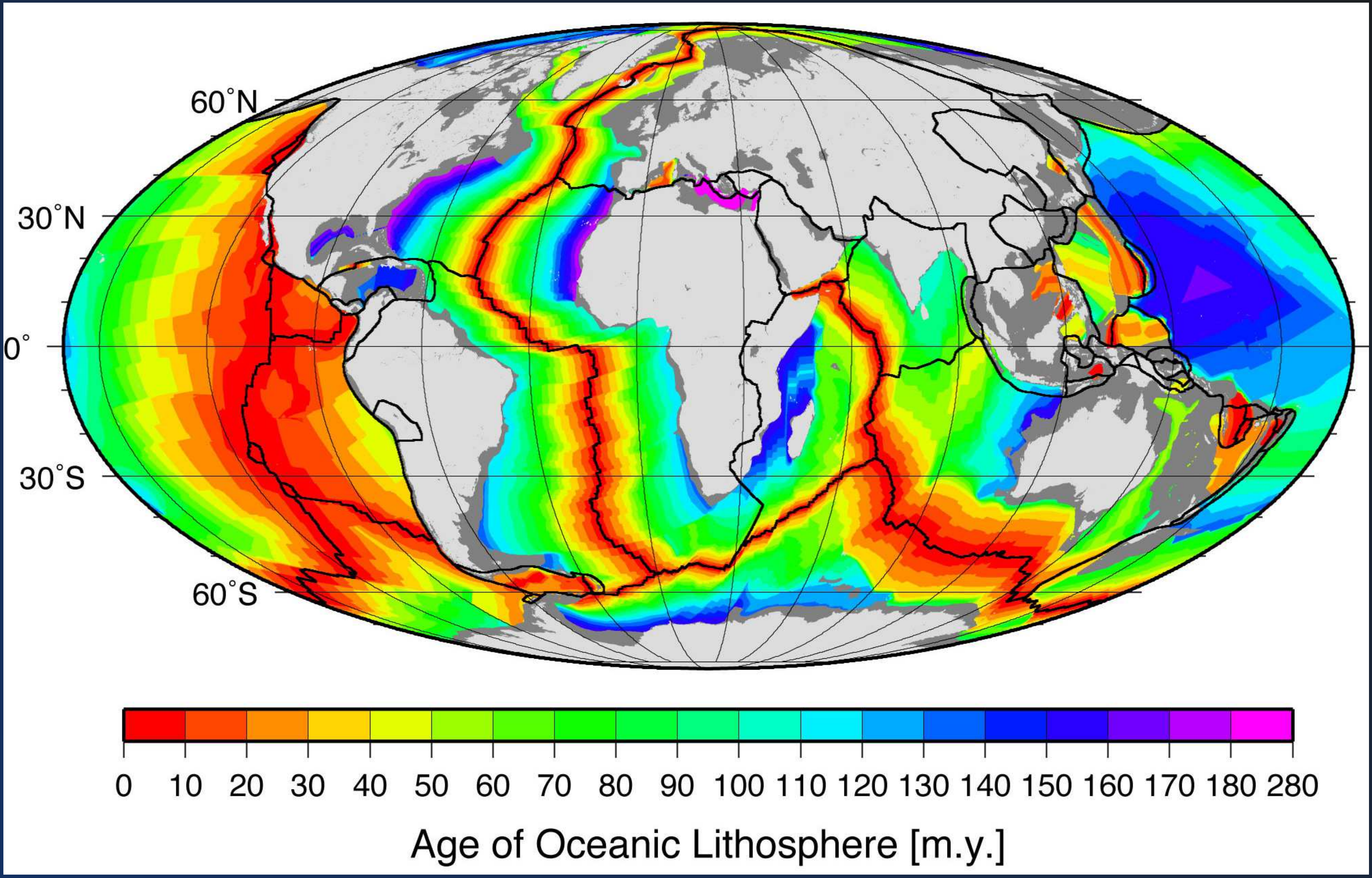


Seafloor spreading

Vine 1966

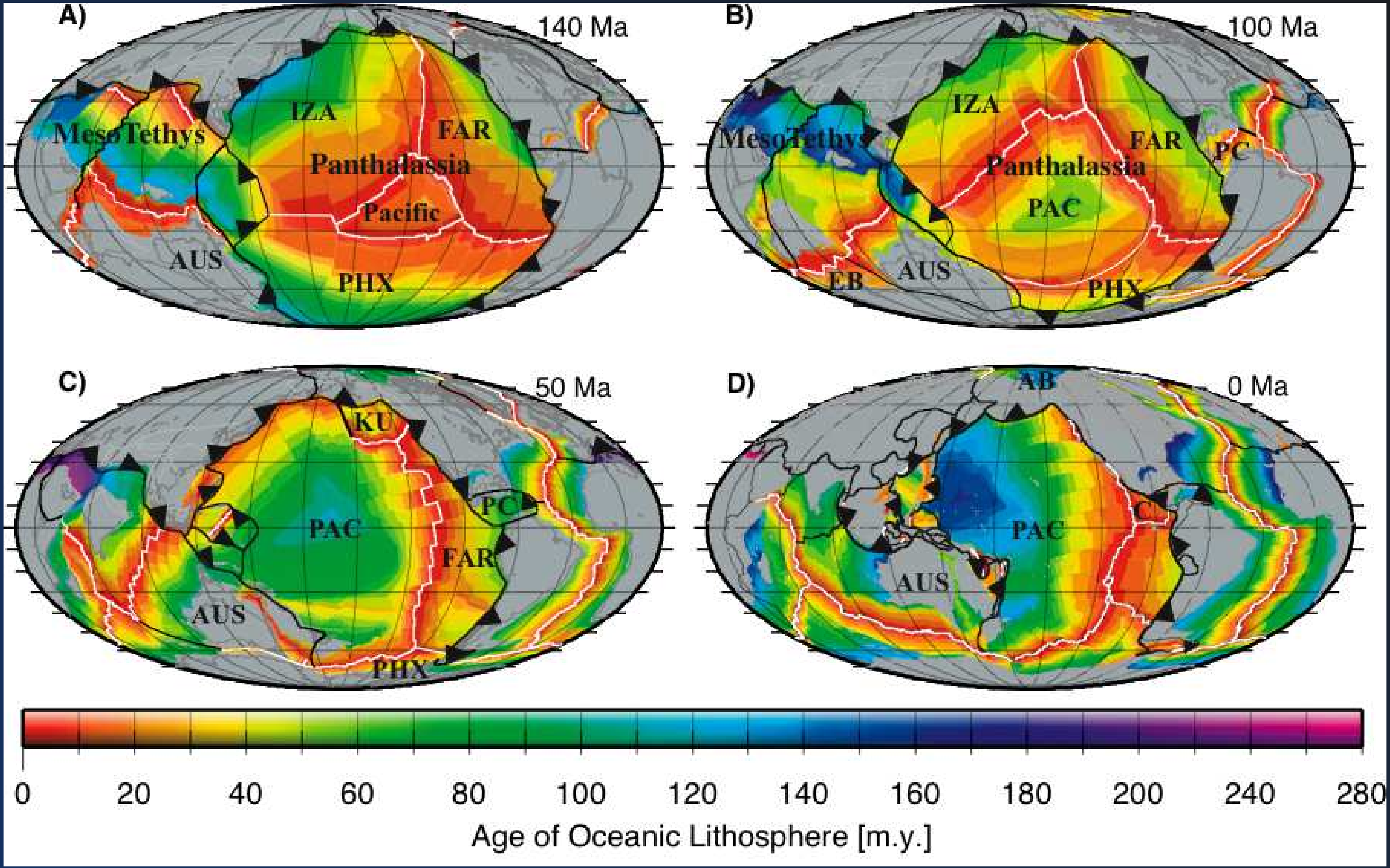


Seafloor Age



Mueller et al. 2008

Seafloor Age



Mueller et al. 2008